Name:

Final Exam Solutions

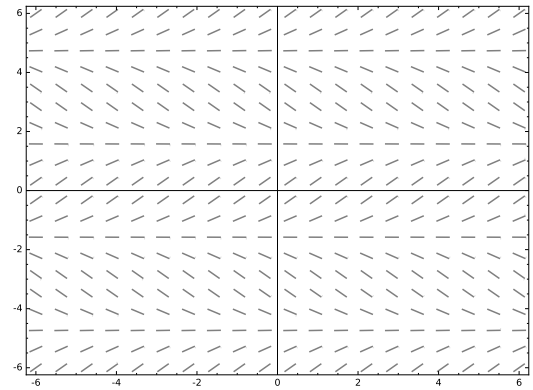
Instructions. The only tools you are permitted to use are pencils, pens, erasers, a sheet of handwritten notes, and your mind. No electronic devices! You have 2.5 hours. Be sure to sign the Honor Code statement on the last page. Good luck!

Problem 1 (True-false, 10 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers. No explanations are required. There will be no revisions for the final exam.

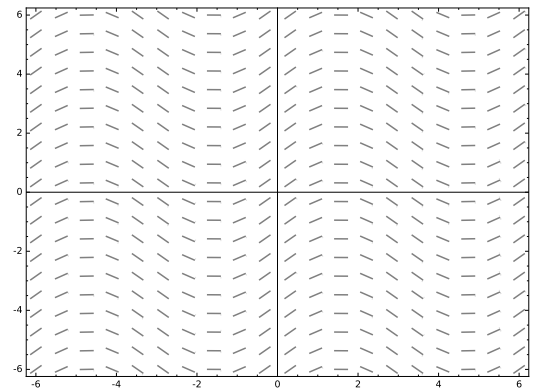
(1)	If A is a matrix with characteristic polynomial $-(\lambda-3)^3$, then A must have 3 linearly independent eigenvectors.	T	**F**
(2)	The functions $\cos(t), \sin(t)$ are linearly independent.	**T**	F
(3)	Suppose $x''' = x - \sin(t)x' + x''$. If $\vec{x} = \begin{bmatrix} x \\ x' \\ x'' \end{bmatrix}$, then $\vec{x}\,' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -\sin(t) & 1 \end{bmatrix} \vec{x}$.	**T**	F
(4)	Every function x satisfying $x' = t - \cos(x)$ must be increasing for all $t > 1$.	**T**	F
(5)	The initial value problem $y' = 2x + \sin(y)$ and $y(0) = 1$ has a unique solution.	**T**	F
(6)	The initial value problem $(y - 2x - 1)y' = 0$ and $y(0) = 1$ has a unique solution.	T	**F**
(7)	Consider the initial value problem $y' = y$ and $y(0) = 1$, whose solution is $y = e^x$. If we run Euler's method with step size 0.2 to approximate $y(1)$, we end up overestimating the true value of $y(1)$.	T	**F**
(8)	Suppose A is a 2×2 matrix of constants and L is the linear operator $L\vec{x} = \vec{x}\,' - A\vec{x}$. The null space of L must be 2 dimensional.	**T**	F
(9)	Suppose A is a 2×2 matrix of constants and L is the linear operator $L\vec{x} = \vec{x}\,' - A\vec{x}$. Then there must exist an $\vec{x}$ such that $L\vec{x} = \begin{bmatrix} e^{2t} \\ \cos(3t) \end{bmatrix}$.	**T**	F
(10)	Suppose $\vec{v}_1, \vec{v}_2$ are linearly independent vectors in $\mathbb{R}^2$. Then there must exist functions g_1, g_2 such that $\begin{bmatrix} \sin(2t) \\ \cos(2t) \end{bmatrix} = \vec{v}_1 g_1(t) + \vec{v}_2 g_2(t)$.	**T**	F

Problem 2 (3 points)**.** On the left are several first order ODEs. On the right are several slope fields. Match each ODE to its slope field.

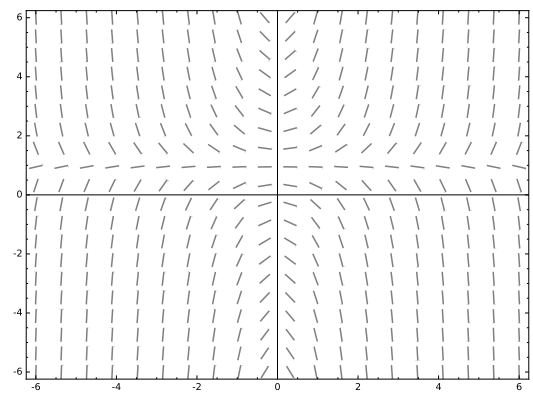
(a) $y' = \cos(x)$



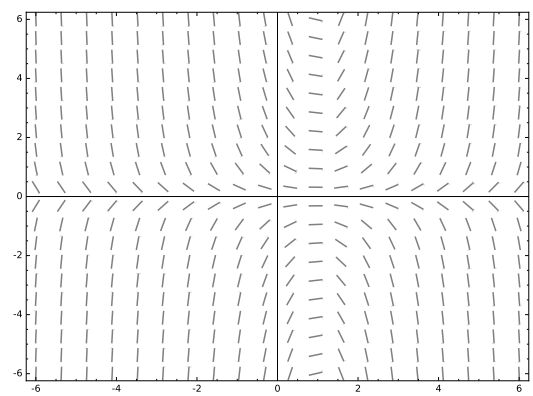
(b) $y' = \cos(y)$



(c) $y' = (x-1)y$



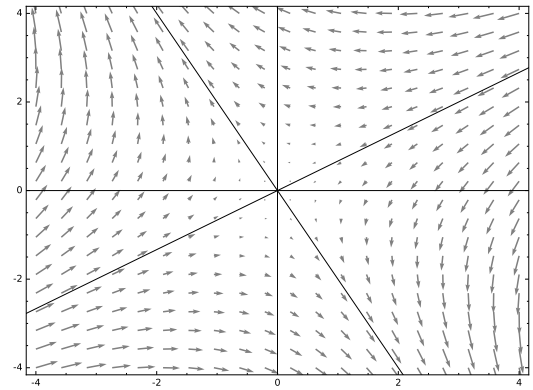
(d) $y' = x(y-1)$



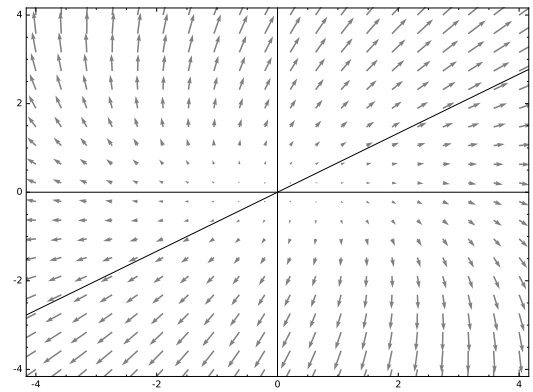
Solution. The slope fields belong to the ODEs (b), (a), (d), (c), respectively.

Problem 3 (3 points)**.** On the left are descriptions of several 2×2 matrices. On the right are several phase portraits. Match each matrix A with the phase portrait of the system $\vec{x}\,' = A\vec{x}$.

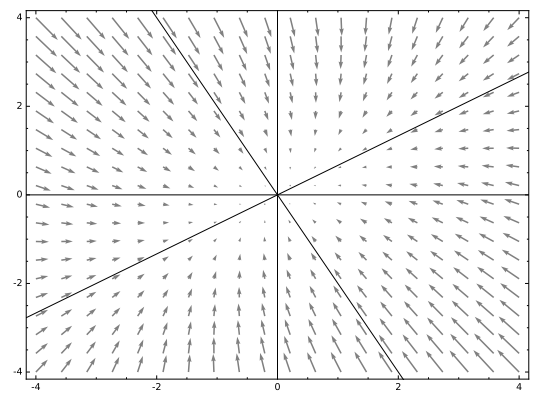
(a) A has complex eigenvalues.



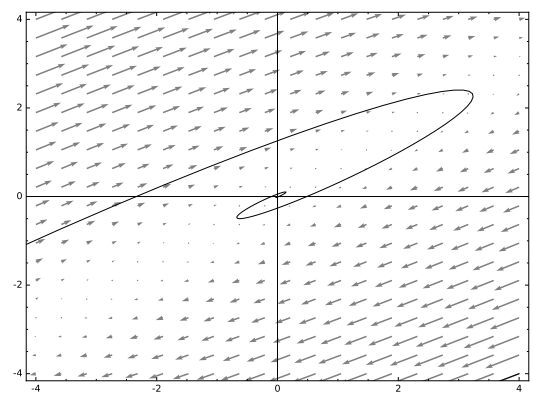
(b) A has negative real eigenvalues.



(c) A has a defective eigenvalue.

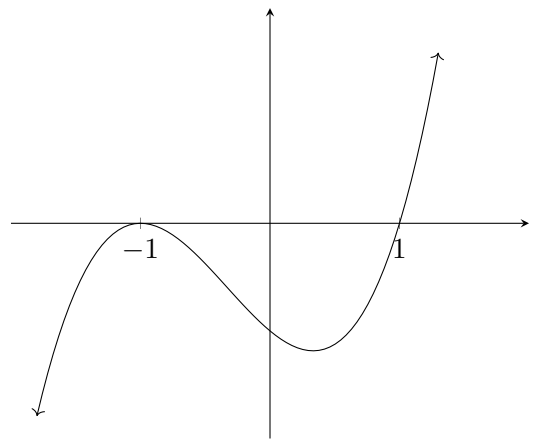


(d) A has two real eigenvalues, one positive and one negative.



Solution. The phase portraits belong to the systems (d), (c), (b), (a), respectively.

Problem 4 (3 points)**.** Consider the autonomous ODE $y' = f(y)$, where f is the function whose graph is depicted below.



(a) Draw a phase diagram. Which, if any, of the critical points are stable?

Solution. The phase portrait looks like this (except rotated 90° counterclockwise).

There are 2 critical points, at $y = -1$ and $y = 1$. Both are unstable.

(b) If y is a solution to the ODE satisfying $y' = f(y)$ the initial condition $y(0) = 0$, what is $\lim_{x\to\infty} y(x)$?

Solution. Following the arrows in the phase diagram, we get $\lim_{x\to\infty} y(x) = -1$.

Problem 5 (2 points)**.** Solve the following initial value problems.

(a) $y' = -2xy^2$ and $y(0) = 1$.

Solution. Note that $y \equiv 0$ is a solution to the ODE, but it does not satisfy the initial condition. Other solutions to the ODE can be found by separating variables.

$$\begin{aligned} -\frac{dy}{y^2} &= 2x\,dx \\ \frac{1}{y} &= x^2 + C \\ y &= \frac{1}{x^2 + C} \end{aligned}$$

Plugging in $x = 0$ and $y = 1$, we find $C = 1$, so the solution is

$$y = \frac{1}{x^2 + 1}.$$

(b) $y' + 3y = 5e^{2x}$ and $y(0) = 2$.

Solution. This equation is linear, so we can use the method of integrating factors. An integrating factor is

$$r(x) = e^{\int 3\,dx} = e^{3x}$$

which means that

$$(e^{3x}y)' = e^{3x}y' + 3e^{3x}y = 5e^{3x}e^{2x} = 5e^{5x}.$$

Antidifferentiating with respect to x, we get

$$e^{3x}y = e^{5x} + C$$

which means that

$$y = e^{2x} + Ce^{-3x}.$$

Plugging in $x = 0$ and $y = 2$, we get $C = 1$, so the solution is

$$y = e^{2x} + e^{-3x}.$$

Problem 6 (2 points)**.** A 1 kg mass is attached to a wall by a spring of stiffness 4 kg/s^2 and slides around on a surface with friction $b > 0$. If $x(t)$ denotes the displacement of the object from equilibrium position at time t, then

$$x'' + bx' + 4x = 0.$$

For what values of b is the system underdamped? Explain briefly.

Solution. The roots of the characteristic equation are

$$r = \frac{-b \pm \sqrt{b^2 - 16}}{2}$$

and the damping depends on the expression under the square root (called the discriminant).

- If $b > 4$, the discriminant is positive and we have overdamping.
- If $b = 4$, the discriminant vanishes and we have critical damping.
- If $b < 4$, the discriminant is negative and we have underdamping.

Problem 7 (2 points)**.** In each of the following, a matrix A is given, together with some information about its eigenvalues. Use the given information to write down the general real-valued solution to the system $\vec{x}\,' = A\vec{x}$.

(a) $A = \begin{bmatrix} 3 & -4 \\ 2 & -1 \end{bmatrix}$ has eigenvalues $1 \pm 2i$, and an eigenvector corresponding to $1+2i$ is $\begin{bmatrix} 1+i \\ 1 \end{bmatrix}$.

Solution. A complex-valued solution to the system is given by

$$e^{(1+2i)t} \begin{bmatrix} 1+i \\ 1 \end{bmatrix} = e^t \begin{bmatrix} \cos(2t) - \sin(2t) + i(\cos(2t) + \sin(2t)) \\ \cos(2t) + i\sin(2t) \end{bmatrix}.$$

Taking real and imaginary parts, we find that the general solution is of the form

$$\vec{x} = c_1 e^t \begin{bmatrix} \cos(2t) - \sin(2t) \\ \cos(t) \end{bmatrix} + c_2 e^t \begin{bmatrix} \cos(2t) + \sin(2t) \\ \sin(2t) \end{bmatrix}.$$

(b) $A = \begin{bmatrix} 5 & -1 \\ 4 & 1 \end{bmatrix}$, $(A - 3I)\begin{bmatrix} 1 \\ 2 \end{bmatrix} = 0$, and $(A - 3I)\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Solution. The information tells us that $(1, 2)$ is an eigenvector with eigenvalue 3 and $(1, 1)$ is a corresponding generalized eigenvector. Thus we have

$$\vec{x} = c_1 e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{3t} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right).$$

Problem 8 (3 points)**.** Suppose that you want to solve the following nonhomogeneous differential equations using the method of undeteremined coefficients. Write down the form of the particular solution that you would guess.

(a) $x'' - 3x' - 10x = e^{5t}$.

Solution. The characteristic polynomial of the associated nonhomogeneous equation is $r^2 - 3r - 10 = (r-5)(r+2)$, so the complementary solution is $x = c_1 e^{5t} + c_2 e^{-2t}$, so the guess should be of the form $x_p(t) = ate^{5t}$.

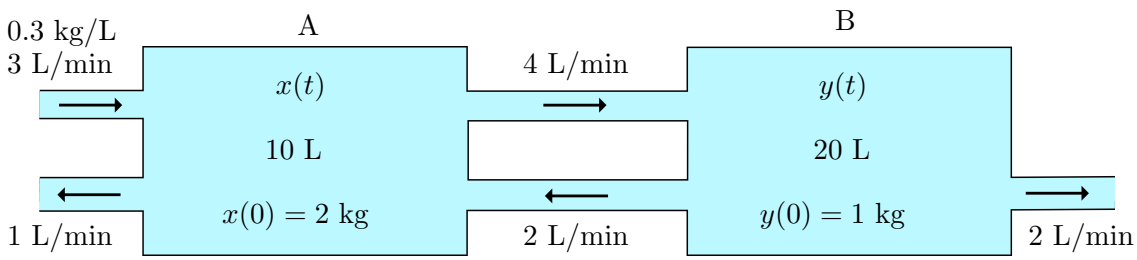
(b) $x'' - 3x' - 10x = t\sin(2t)$.

Solution. $x_p(t) = (at+b)\sin(2t) + (ct+d)\cos(2t)$.

(c) $\vec{x}\,' = \begin{bmatrix} 2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix} \vec{x} + \begin{bmatrix} e^{2t} \\ 5 \\ e^{4t} \end{bmatrix}$

Solution. $\vec{x}_p(t) = \vec{a}te^{2t} + \vec{b}e^{2t} + \vec{c} + \vec{d}e^{4t}$

Problem 9 (3 points)**.** Suppose there are two interconnected tanks A and B with salt water flowing between them as depicted in the diagram below. Suppose further that the input coming into tank A from outside has a constant salt concentration of 0.3 kg/L. Let $x(t)$ and $y(t)$ denote the quantities of salt that are dissolved in tanks A and B at time t, respectively. Assume perfect mixing.



(a) Find the matrix A and the vector-valued function $\vec{f}$ such that

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = A\begin{bmatrix} x \\ y \end{bmatrix} + \vec{f}.$$

Solution. We have

$$\frac{dx}{dt} = -\frac{5x}{10} + \frac{2y}{20} + 0.9 \quad \text{and} \quad \frac{dy}{dt} = \frac{4x}{10} - \frac{4y}{20}.$$

Thus

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} -1/2 & 1/10 \\ 2/5 & -1/5 \end{bmatrix}}_{A}\begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} 0.9 \\ 0 \end{bmatrix}}_{\vec{f}}$$

(b) What is $\lim_{t\to\infty}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$?

Solution. One expects that, over time, both tanks will end up with a salt concentration of 0.3 kg/L. This means that tank A will have 3 kg of salt and tank B will have 6 kg of salt, so

$$\lim_{t\to\infty}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Problem 10 (Extra credit! 2 points, no partial credit)**.** Ignoring constants of integration, find all functions f such that

$$\left(\int f(x)\,dx\right)\left(\int \frac{1}{f(x)}\,dx\right) = -1.$$

Solution. There are many approaches. One possibility is to take the above equation, differentiate several times until you reach a differential equation you can solve. You should find that $f(x) = ce^{\pm x}$ for any nonzero constant c.

Honor code. If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

This page has been left blank for scratchwork.

This page has been left blank for scratchwork.